current created in the stator windings to direct current usable for recharging your batteries.

- *Voltage regulator.* This critical device adjusts the current in the field windings to match your alternator's output and the needs of your batteries. The voltage regulator also controls the output of the alternator so that it doesn't overcharge your batteries.

Alternators work on the principle of *induction* (discussed in the Word about Inductive Pickups sidebar in chapter 3) whereby a magnet placed adjacent to a wire will induce electrical current in the wire. Each stator winding in your alternator has a very long wire wound in a coil, and the rotor is a powerful electromagnet. By spinning the rotor inside the stator, electrical current is induced in the stator, and by varying the amount of magnetism in the rotor (the primary function of the voltage regulator) the amount of electrical current produced by the stator can be precisely controlled. Remember, all conductors with electrical current flowing through them are surrounded by a magnetic field. We can induce electron flow in nearby conductors by rapid movement of this field in proximity to the conductors (this works in reverse as well, by moving a conductor rapidly in close proximity to a stationary magnet).

If the stator windings in your alternator (there are usually three) all had the same polarity (the positive and negative terminals arranged the same way), you wouldn't have an alternator, you would have a generator producing direct current. However, alternators have their stator windings arranged in fingers so that as the rotor rotates past the stator windings, the positive and negative poles are constantly and rapidly reversing, thus inducing alternating current. Because your battery needs to be charged with direct current, this alternating current is passed through a bridge rectifier where it's converted to usable direct current.

Why, you may well ask, do we go to all this trouble to generate alternating current, and then convert it back to direct current, when an old-fashioned generator would develop direct current without all the complicated circuitry of the bridge rectifier? The answer is efficiency. An alternator can be built much lighter than a generator, and it will produce much higher currents than a generator of the same size. It's ironic, but true: it's easier and better to produce alternating current and convert it to direct current than it is to produce direct current straight from a generator.

Engine-Driven Marine Alternators

There are four broad categories of engine-driven charging systems commonly found on boats today. The chances are better than good that your boat's charging system will fall into one of the following groups.

1. Outboard engine with remote rectifier and no voltage regulator
2. Outboard engine with remote rectifier and voltage regulator
3. Inboard or stern drive with remote voltage regulator
4. Inboard or stern drive with integral voltage regulator

Alternator Problems

Most problems with alternators are best corrected by removing the alternator from the boat and taking it to an alternator shop, which has the special equipment needed to make delicate repairs. Some manufacturers have factory exchange programs that allow you to exchange your unit for an identical factory-rebuilt replacement—for a fee, of course. It's a simple matter to pop the old alternator out, exchange it at the dealer's, then pop the new one back in. This exchange is sometimes available no matter what condition your old alternator is in—a real blessing when you experience an engine fire or after your alternator gets doused with salt water. Many offshore boats carry a spare rebuilt alternator that can be substituted for a defective unit if it should give up the fight while fishing the Gulf Stream or on a long weekend in the Bahamas.

This swap-and-replace procedure presupposes that you're able to diagnose when and why the alternator in your system is acting up.